

Pragmatic Effects in LLM Prompting

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Basics about Language Models

Modern language models are *causal*. This means that, unlike BERT of the past, these models only look in one direction—forward.

The model has an internal state, predicting the value of the next token, conditioned on all the past tokens.

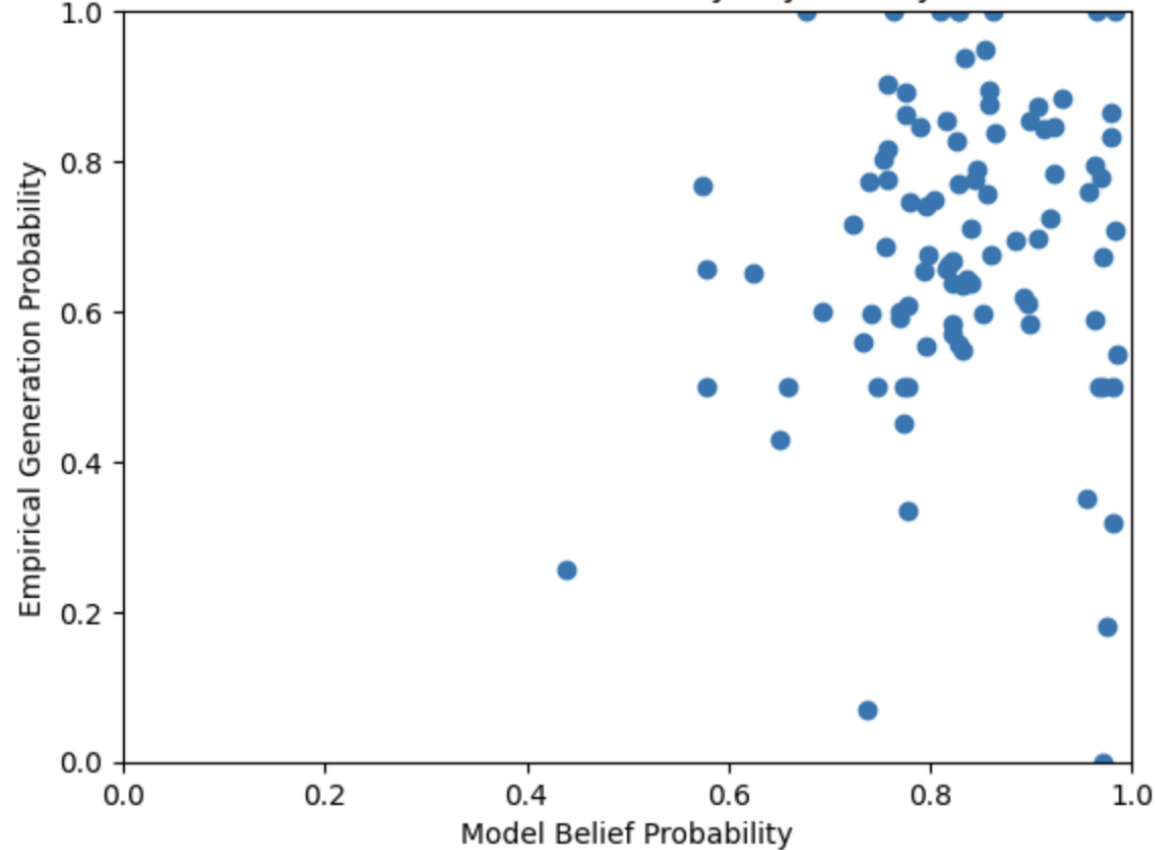
It also has a generation head, which adds stochasticity to the internal state and generates the next token based on the past tokens it has generated, acting sort of like a state-space machine.

Basics About Linguistic Meaning

- The meaning of an utterance is ascertained by its *semantics*
 - Classically, this has been on the basis of a sentence's truth-conditions
 - In language models, it is on the basis of a sentence's place in statistical distributions
- Pragmatic effects (such as inflect, intonation, framing) can lead two semantically equivalent sentences to have completely different meanings
 - e.g.
 - I didn't say she stole the money → Someone else said it.
 - I *didn't* say she stole the money → Denial of having said it.
 - I didn't say she stole the money → Maybe implied it, but didn't state it.
 - I didn't say *she* stole the money → Someone else might have stolen it.
 - I didn't say she *stole* the money → She might have borrowed it.
 - I didn't say she stole the *money* → She stole something else.
- The goal of this work is to study how structural, contextual, and interactional features of language presentation modulate meaning beyond literal semantics
 - As an area which has historically shown to be concerning in these model, we test these w.r.t meanings built around gender in the model to see if new supposed “improvements” are robust to these modulations of meaning

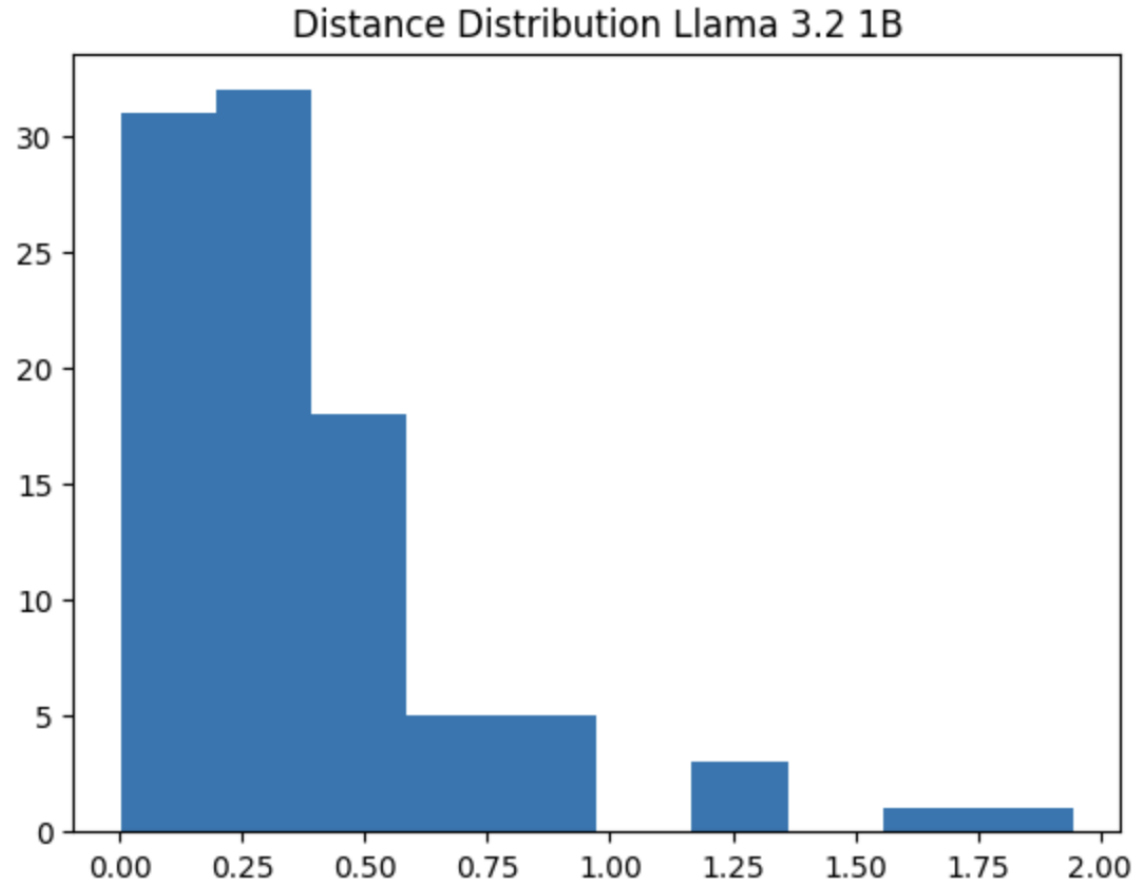
Belief vs. Generation

Llama 3.2 1B T=0.5 Female Pronoun Probability Asymmetry (No Context Sentence)



Model beliefs generally make female pronoun more probable than male. Generation, however is all over the place.

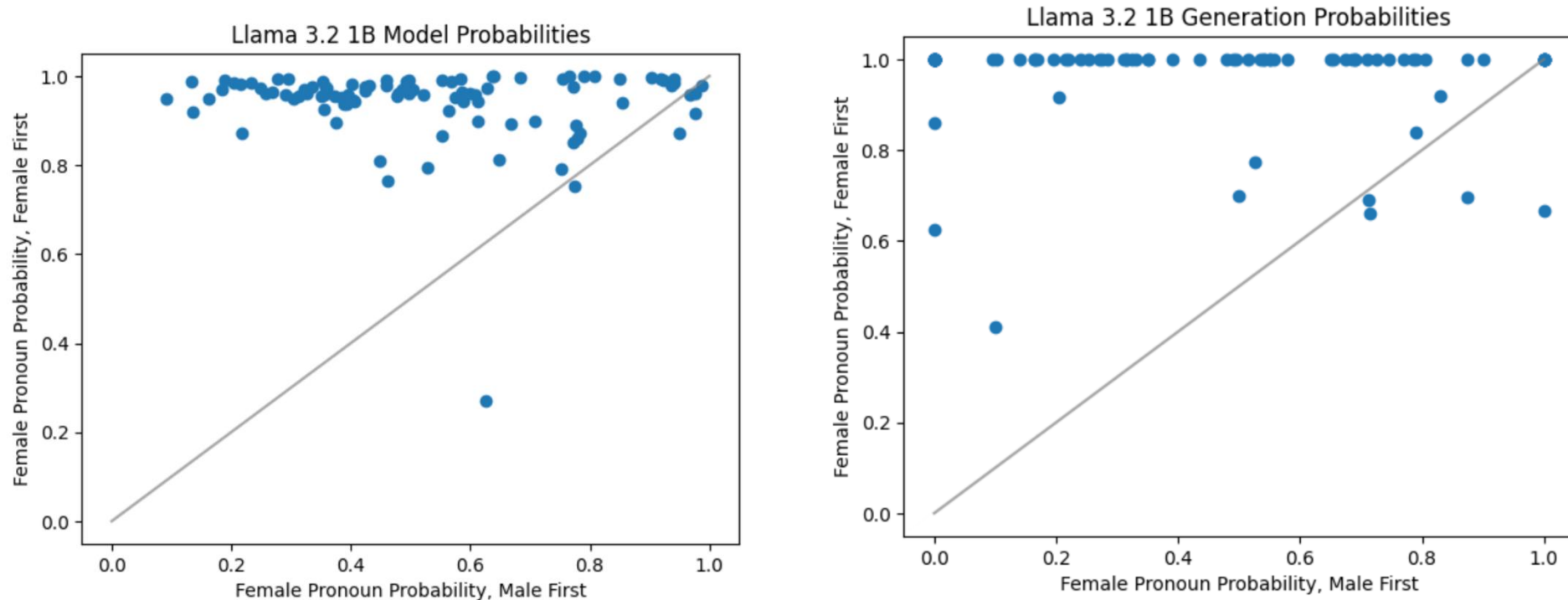
Belief vs. Generation



This is the distribution of the L1-distance between the internal probabilities and the generated probabilities.

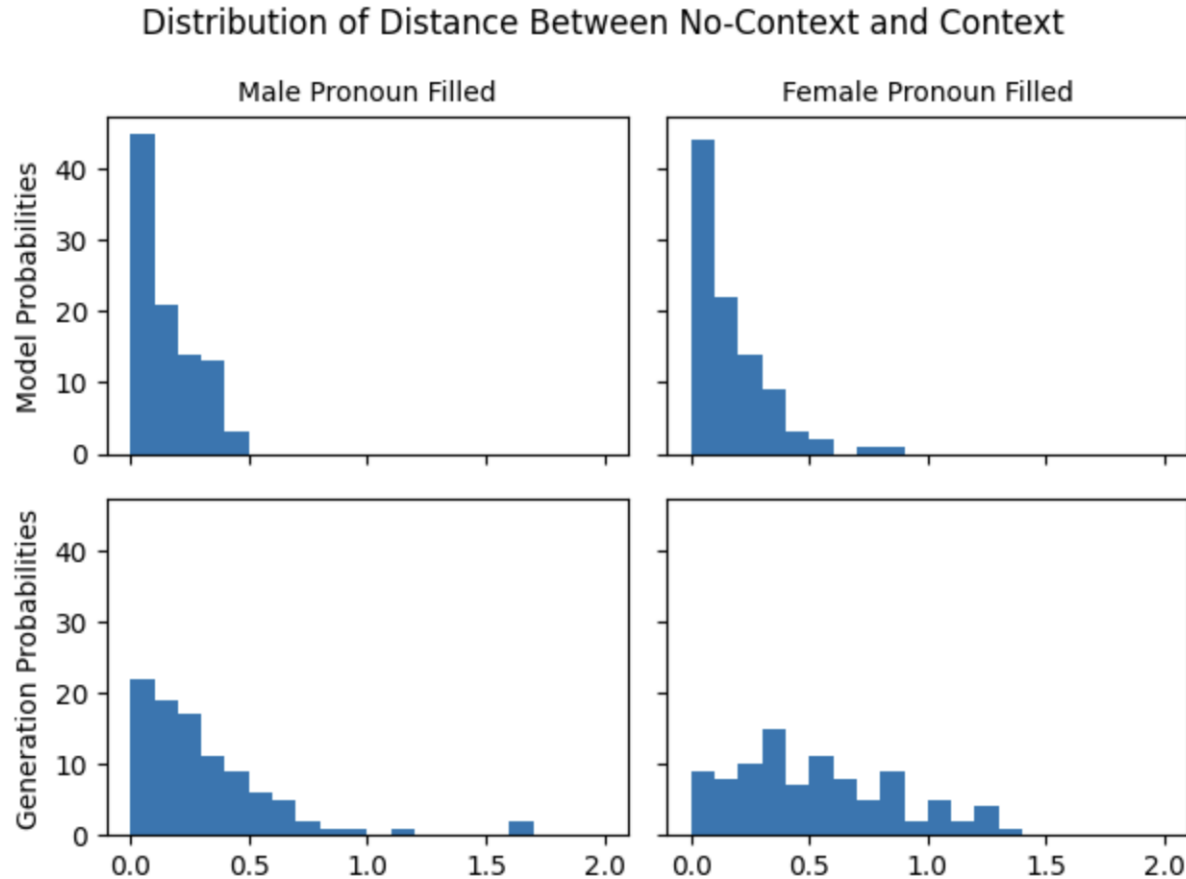
The peak on the left suggests that most items in the Winogender dataset stay consistent between the two, but there are some instances where are nearly maximally different at a temperature of only 0.5.

Ordering Effects



In **both** the model beliefs and in generation, using the female pronoun **all but guarantees** the female pronoun will be chosen, *regardless of the utterance*.

Priming Effects



In every instance in the dataset, there is are two characters. In this test, we provide a “priming” sentence which provides a gendered pronoun for the other character. Hypothetically, this should be irrelevant to the resolution of the pronoun requested. Here, we show the L1-distance between the primed (context) and un-primed (no context) prompts.

The differences are increased when the priming sentence is filled with a female pronoun. This is especially pronounced for generation.

Framing Effects

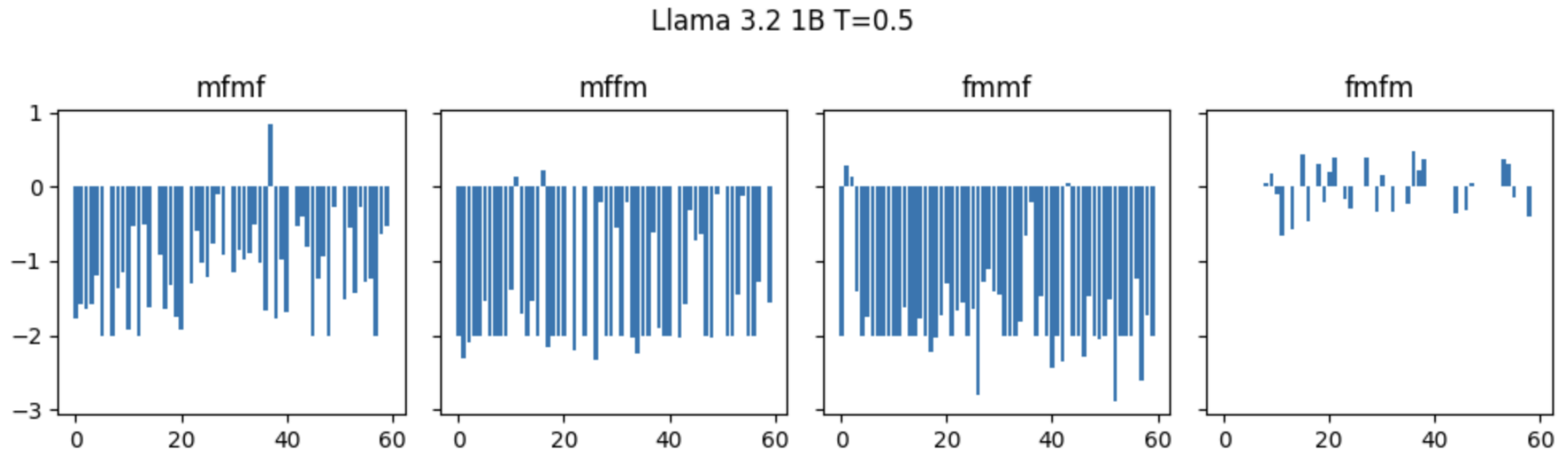
- Naming game framing—very much follows the cooperative principles of Grice and Habermas

Results currently being collected

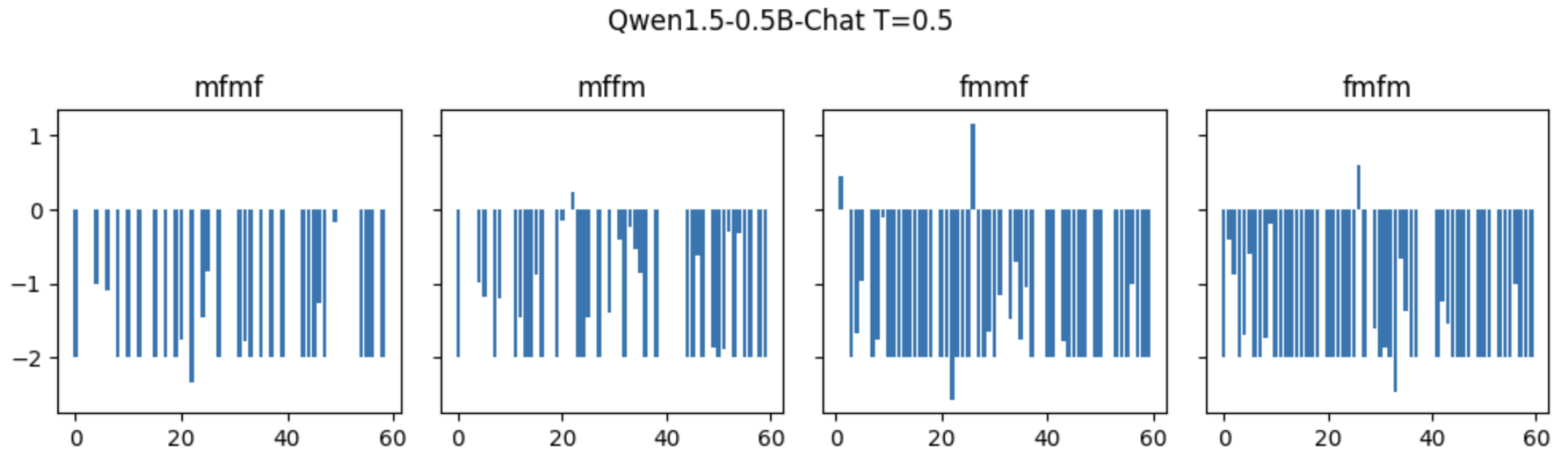
Contextuality

- How do we know if we are, in fact, measuring the same construct/factor/dimension when we get different values from all of these effects? → *Contextuality*
- Contextuality requires *at least two measurements* can be interpreted in several ways.
- If we ask a model to resolve the pronoun for sentence A and the pronoun for sentence B and we find that this sentence is *contextual*, it means:
 - There is (logically) no way we can create a model of B which does not depend on A
 - In LLMs, we might think of this as saying that there is no “latent dimension” for B which is not a function of A
 - For gendered pronouns, this would be to say that if the model resolves the pronoun “her,” for B this is not coming from some “gender axis” but rather, is a function of the pronoun it ascribed A
 - In psychology, this is the “selectiveness of influences” which answers the question of whether, logically, the measurements can be attributed to the same latent construct

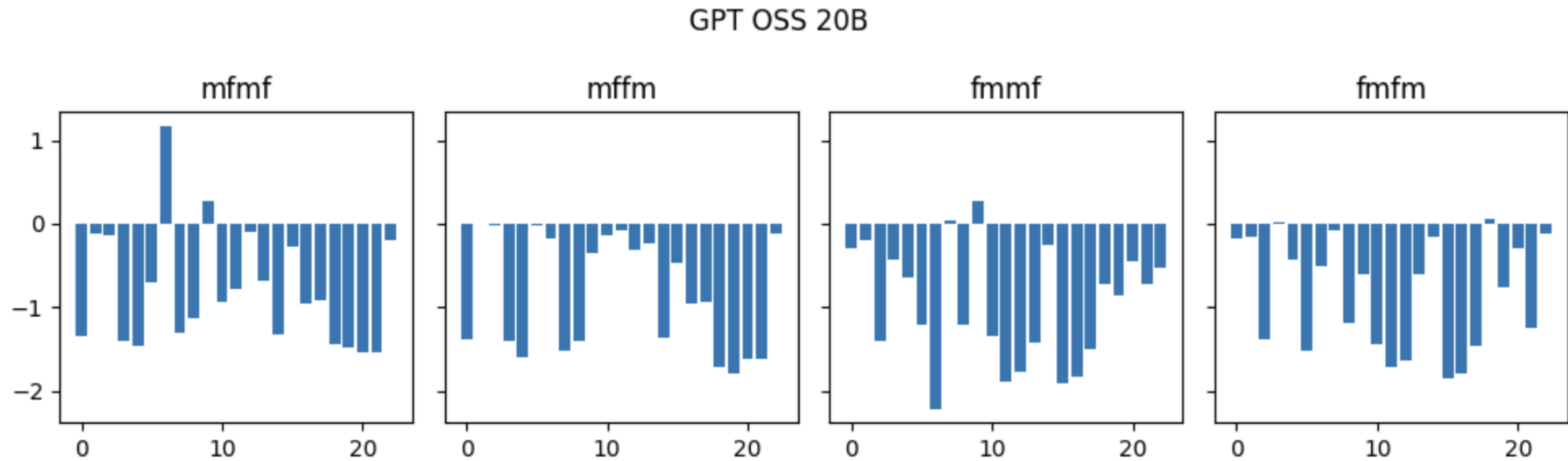
Contextuality in Llama 3.2 1B



Contextuality in Qwen 1.5 0.5B



Contextuality in GPT-OSS 20B



Takeaways

- A two part story is forming:
 - testing pragmatic effects (order, priming, and framing)
 - determining whether the meaning of “he/she/his/her/etc.” is stable
 - Some difficulties trying to work out how to explain this and its consequences, so a conversation would be good
- So far, there seem to be important results w.r.t gender in prompting.
 - Female-pronoun-first options lead to heavy skews in the responses
 - Female character in priming sentences lead to skews especially in generation
 - *TODO: Move from aggregate assessment to how this aligns with our morals and statistics*
 - *i.e. maybe it's a good thing that the model is more prone to refer to the CEO as a woman*